Data Quality Report (OWID Energy Dataset)

1. Dataset Overview

Title

The dataset selected for this project is the *Our World in Data (OWID) Energy Dataset*.
https://github.com/owid/energy-data

What the Dataset Contains

This dataset provides a comprehensive collection of global energy-related statistics across countries and over time. It includes:

- Total and per-capita energy consumption
- Electricity generation by source (coal, oil, gas, nuclear, solar, wind, hydro, etc.)
- Energy mix shares (renewables vs fossil fuels)
- CO_2 and greenhouse gas emissions

Why This Dataset Was Selected

- The dataset is *Findable* (hosted on GitHub), *Accessible* (free CSV download), *Interoperable* (standard format), and *Reusable* (clear licensing and documentation).
- Energy and climate data are central to global policy and sustainability discussions, making this dataset highly meaningful.
- It contains realistic issues such as missing values, mixed entity types, etc.
- Large enough to reflect real-world complexity while still being manageable for analysis.

2. Main Problems Identified

During the initial profiling stage in Tableau Prep, the following issues were identified:

- **Missing Values (Completeness):**
  Several variables like renewable energy indicators, contained a high proportion of null values. This was particularly noticeable for earlier years and smaller or developing countries.
- **Non-Country Aggregates (Validity):**
  The dataset included entries such as *World*, *OECD*, *G20*, and regional groupings (e.g., *Africa (EI)*). Including these alongside countries would distort analysis through double counting.
- **Excessive and Partially Irrelevant Columns (Relevance):**
  Although reduced from the original dataset, the working dataset still contained **56 columns**, not all of which are equally relevant for a focused country-level analysis.

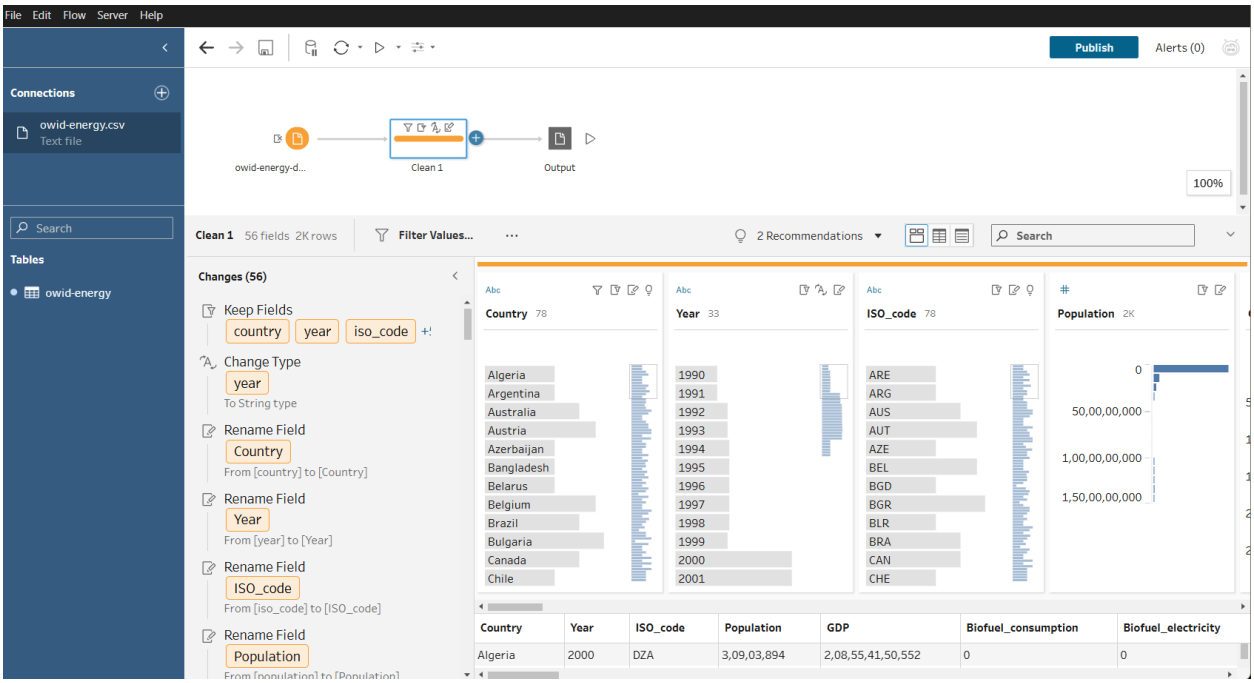
- **Incorrect Data Type (Validity):**
  The *year* field was interpreted as a decimal number (e.g., 1,900), which is not suitable for categorical or temporal analysis.
- **Inconsistent Naming (Consistency):**
  Column names were initially in lowercase, making them less intuitive for reporting and business use.
- **Lack of Built-in Validation Checks:**
  There was no mechanism to verify whether energy shares (renewables vs fossil fuels) were logically consistent.
- **Potential Outliers:**
  Some extreme values were present for small countries or territories, which could skew aggregated insights.

3. Steps Applied in Tableau Prep

All transformations were implemented within a structured Tableau Prep workflow, ensuring traceability.

- **Data Loading & Profiling:**
  The dataset was imported and explored using Tableau Prep's profiling pane to identify nulls, distributions, and inconsistencies.
- **Column Selection (Relevance):**
  The dataset was reduced from ~130 columns to **56 relevant columns**, balancing completeness with usability. Instead of aggressively filtering, a broader set of variables was retained to preserve analytical flexibility.
- **Filtering Non-Country Entities (Validity):**
  Regional and aggregate entities (e.g., *World*, *OECD*, *G20*, *Europe (Ember)*) were excluded to ensure only individual country-level records remained.
- **Fixing Data Types (Validity):**
  The *year* column was converted from a decimal format to **string**, eliminating formatting issues and preventing incorrect numerical interpretation.
- **Renaming Columns (Consistency):**
  All selected columns were renamed into **clear, readable labels**, including units where applicable. This significantly improved interpretability.
- **Creating a Validation Field (Validity):**
  A calculated field (*Clean_energy_share_pct*) was created to validate whether renewable and fossil electricity shares logically sum up.
- **Handling Missing Values (Completeness):**
  Rows with missing values in the **core metric (primary_energy_consumption)** were removed. Other missing values were retained, as they represent genuine data gaps.
- **Exporting Clean Data:**
  The cleaned dataset was exported as a CSV file, ready for further analysis.

4. Tableau Prep Screenshot



5. Conclusion

The data cleaning process significantly improved the quality and usability of the dataset by making it more structured, consistent, and reliable for analysis. Key issues such as irrelevant columns, non-country entities, and incorrect data formats were addressed, resulting in a cleaner and more focused dataset.

At the same time, some limitations—such as missing values and the presence of outliers—remain, reflecting real-world data challenges rather than errors. Overall, the dataset is now well-prepared for analysis, with improved transparency and a reproducible workflow, while still requiring mindful interpretation of its remaining gaps.